# MOCO: Measure Once, Compile Once

## Fraud detection on a quantum budget

| | |
|---|---|
| **Measure once** | 2 circuit executions per training transaction, in total |
| **Train convexly** | global optimum in PyTorch, no circuit inside the training loop |
| **Noise cannot change the ranking** | Spearman ρ = 1 at every noise level, by construction |

Darius Tatomir · Deliu Georgiana | BreaQ 2026, Quantum AI track, Task 2

# The baseline breaks under noise: AUPRC 0.580 to 0.405

**492**

frauds in 284,807 transactions (0.17%)

- A model that flags nothing is 99.83% accurate: accuracy says nothing. We rank by AUPRC (precision-recall).
- Common split, stratified 60/20/20, seed 42: test 56,962 rows with 98 frauds; threshold chosen on validation at the same noise level.

**Baseline audit (kit VQC: ZZ map + EfficientSU2)**

**93**
gates

**33**
CNOT

**45**
depth

**126**
noise locations

2,859,360 circuit executions to train and validate

**Test AUPRC 0.580 ± 0.053 at p = 0 → 0.405 ± 0.177 at p = 0.05 (5 seeds)**

# Two facts from Task 1 make noise harmless for the ranking

### Exercise 3.2: squashing keeps AUPRC

AUPRC depends only on the order of the scores. Compressing them towards 1/2 keeps the order, so AUPRC and AUROC do not change; only the threshold has to move.

$$s \;\to\; 1/2 + c\,(s - 1/2), \quad c > 0$$

### Exercise 10.1: depolarizing scales ⟨Z⟩

A depolarizing channel after a one-qubit gate shrinks the Bloch vector by the same factor for every input.

$$\langle Z \rangle \;\to\; (1 - 4p/3)\,\langle Z \rangle$$

**MOCO = a circuit where every qubit sees the same number of one-qubit gates and no CNOT, so the whole score is squashed by one factor $\lambda^2$ with $\lambda = 1 - 4p/3$.**

# Every choice is made on train or validation, never on test

- Encoder fitted on the training subset only: scale Time and Amount, PCA to 4 components, angles in [0, π].
- Undersample the training split, keeping all 295 training frauds; ratio and feature map chosen on validation (single 1:10: validation AUPRC 0.675 ± 0.012, 5 seeds).
- C of the convex fit chosen by 5-fold CV inside the training subset: no validation executions.
- Test set touched once, at the end; threshold maximises F1 on validation at the same noise level.

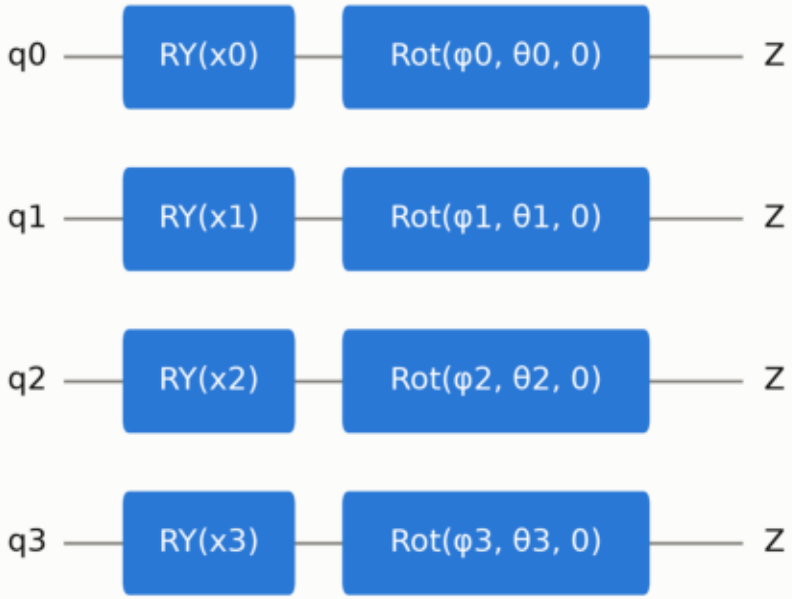
| feature map | ratio | val AUPRC (5 seeds) | train exec. |
|---|---|---|---|
| single | 1:1 | 0.564 ± 0.098 | 1,180 |
| single | 1:3 | 0.663 ± 0.020 | 2,360 |
| single | 1:10 | 0.675 ± 0.012 | 6,490 |
| single | 1:30 | 0.673 ± 0.018 | 18,290 |
| single | 1:100 | 0.648 ± 0.015 | 59,590 |
| pair | 1:1 | 0.623 ± 0.017 | 1,770 |
| pair | 1:3 | 0.645 ± 0.019 | 3,540 |
| pair | 1:10 | 0.663 ± 0.011 | 9,735 |
| pair | 1:30 | 0.676 ± 0.016 | 27,435 |
| pair | 1:100 | 0.628 ± 0.015 | 89,385 |

**Chosen: single 1:10, the cheapest within one standard error of the best mean (pair 1:30, 0.676, at 4.2x the executions).**

# MOCO has 8 noise locations, the baseline 126

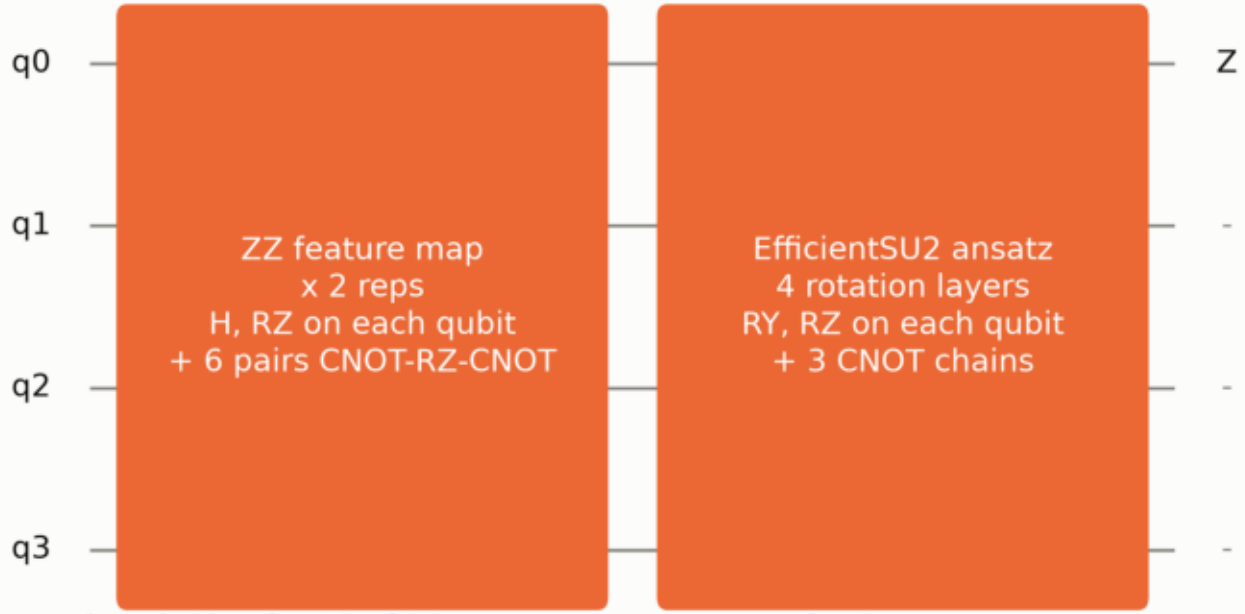
Every gate is followed by a depolarizing channel on each qubit it touches: 8 vs 126 noise locations

MOCO: 8 gates, 0 CNOT, 8 noise locations, depth 2




readout H = Σ w_q Z_q, p_fraud = (1 − ⟨H⟩) / 2

Baseline: 93 gates, 33 CNOT, 126 noise locations, depth 45




readout ⟨Z0⟩ only; 12 of 32 parameters never reach q0

# Training the circuit is a convex fit on data measured once

## 1 Measure once

RY(x)|0⟩ has Bloch vector (sin x, 0, cos x). Read every training transaction in the X and Z bases: 2 executions per row, 6,490 in total.

## 2 Heisenberg picture

After a local Rot, $\langle Z_q \rangle = m_q \cdot r_q(x)$ exactly. The readout $H = \Sigma\, w_q Z_q$ is linear in the measured Bloch vectors.

## 3 Convex fit

Logistic regression on the measured vectors in PyTorch (LBFGS), C by CV on train: global optimum, no barren plateau, no circuit in the loop.

## 4 Compile back

Coefficients become Rot(−φ, −θ, 0) angles and observable weights; $p_{fraud} = (1 - \langle H \rangle)/2$. The circuit does the classification.

*Same idea as classical surrogates of quantum models (Schreiber, Eisert, Meyer, PRL 2023) and data acquisition followed by classical processing (Cerezo et al., 2025): here the result is compiled back into the circuit.*

# Noise only squashes MOCO's scores: the ranking cannot change

1 Each depolarizing channel on a qubit shrinks its Bloch vector by $\lambda = 1 - 4p/3$ (exercise 10.1).

2 No CNOT and k = 2 gates on every qubit: the state stays a product, every $\langle Z_q \rangle$ is multiplied by $\lambda^2$.

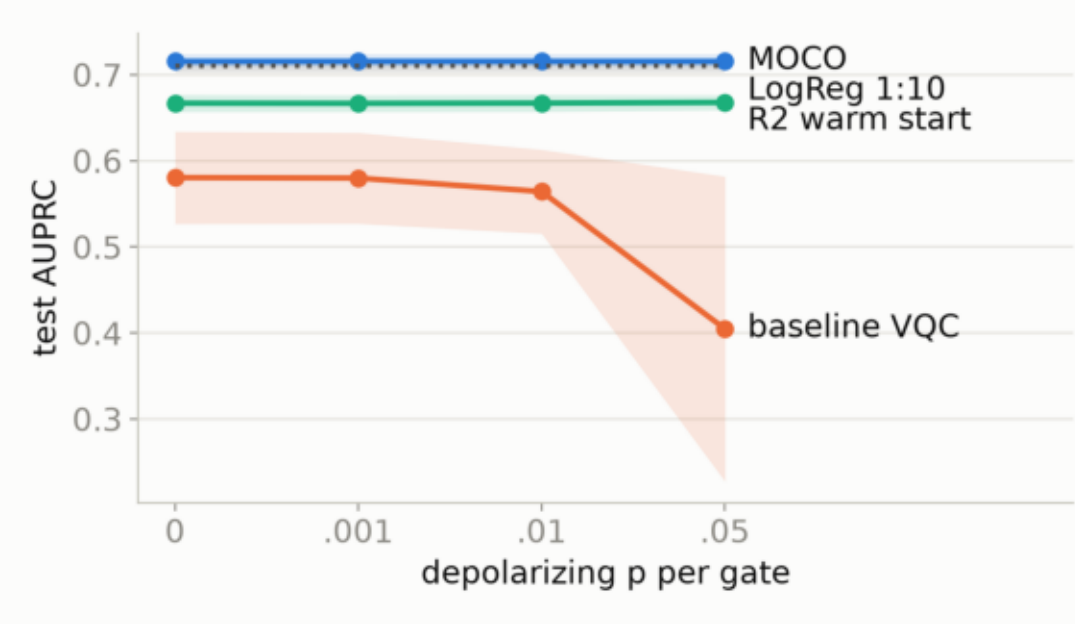
3 The readout is linear, so the whole score is squashed by $\lambda^2$ around 1/2 (exercise 3.2).

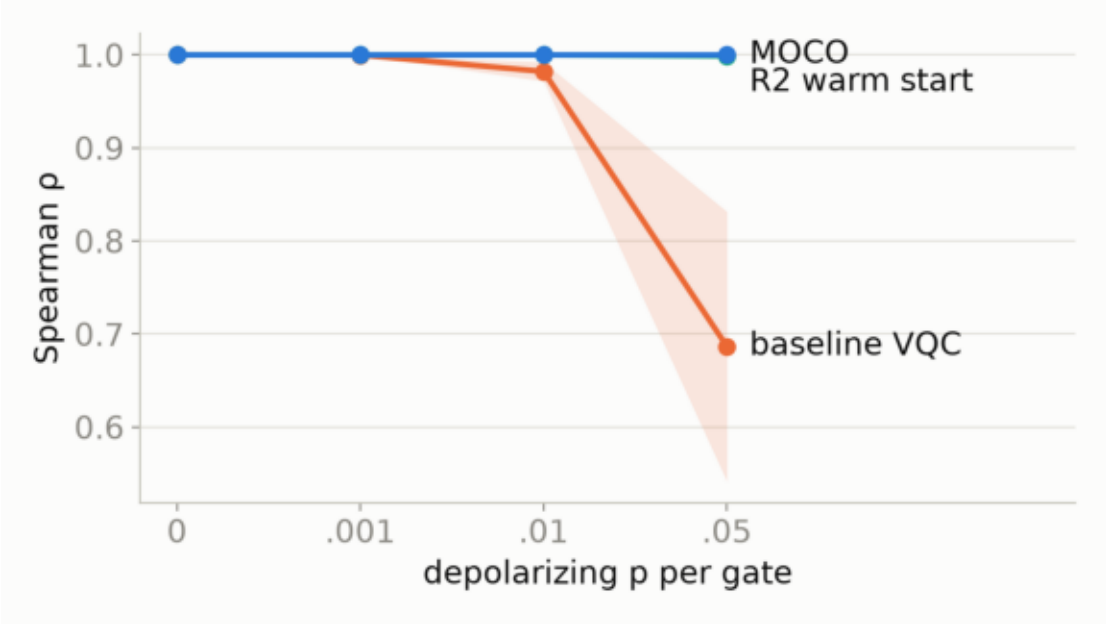
$$s_{noisy}(x) = 1/2 - \lambda^2 (1/2 - s_{ideal}(x))$$

- AUPRC, AUROC and $\rho = 1$ identical at every $p < 3/4$; the max-F1 threshold moves analytically: $t_p = 1/2 - \lambda^2 (1/2 - t_0)$.
- Checked against the kit's noise model (noisy_predict): max error 5.7e-14 on 200 rows at p = 0.05 and 0.5; effective compression equals $\lambda^2$ to 3e-17.

# MOCO matches logistic regression and stays flat under noise

| model (5 seeds) | AUPRC p=0 | p=0.001 | p=0.01 | p=0.05 | ρ .05 | F1 .05 | MCC .05 |
|---|---|---|---|---|---|---|---|
| baseline VQC | 0.580 ± 0.053 | 0.580 ± 0.053 | 0.564 ± 0.049 | 0.405 ± 0.177 | 0.686 | 0.537 | 0.545 |
| R2 random init | 0.635 ± 0.018 | 0.635 ± 0.018 | 0.636 ± 0.017 | 0.638 ± 0.015 | 0.999 | 0.760 | 0.760 |
| R2 warm start | 0.667 ± 0.010 | 0.667 ± 0.010 | 0.667 ± 0.010 | 0.668 ± 0.009 | 0.999 | 0.798 | 0.798 |
| **MOCO** | **0.716 ± 0.007** | **0.716 ± 0.007** | **0.716 ± 0.007** | **0.716 ± 0.007** | **1.000** | **0.800** | **0.801** |
| LogReg 1:10 | 0.710 ± 0.007 | 0.710 ± 0.007 | 0.710 ± 0.007 | 0.710 ± 0.007 | - | 0.794 | 0.795 |
| LogReg 1:1 | 0.703 ± 0.010 | 0.703 ± 0.010 | 0.703 ± 0.010 | 0.703 ± 0.010 | - | 0.791 | 0.791 |



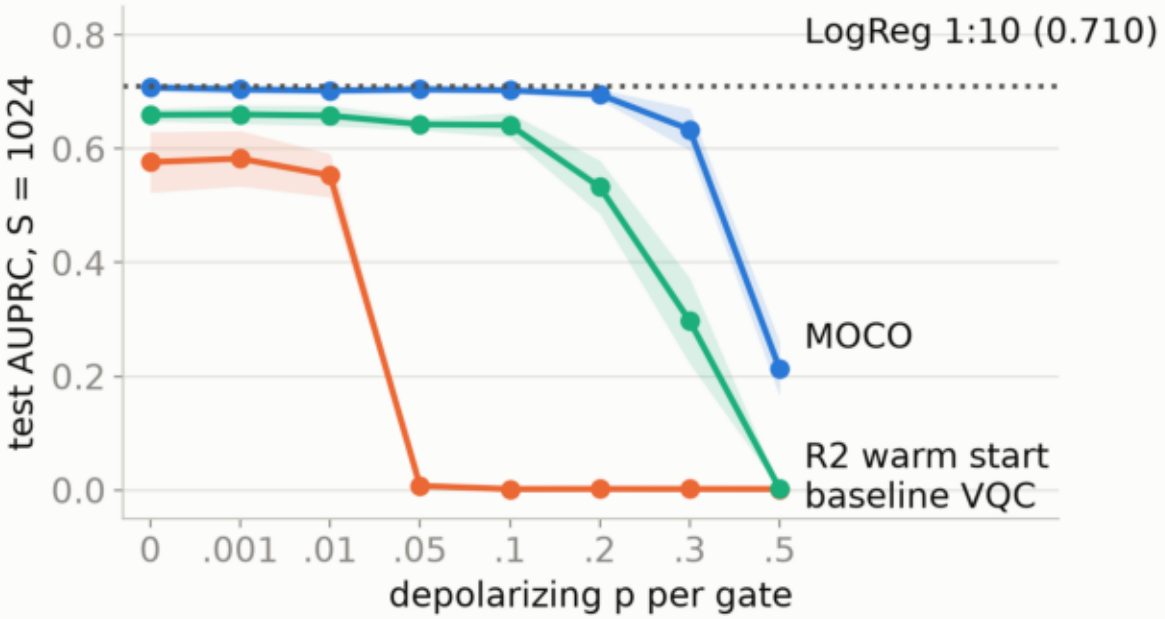



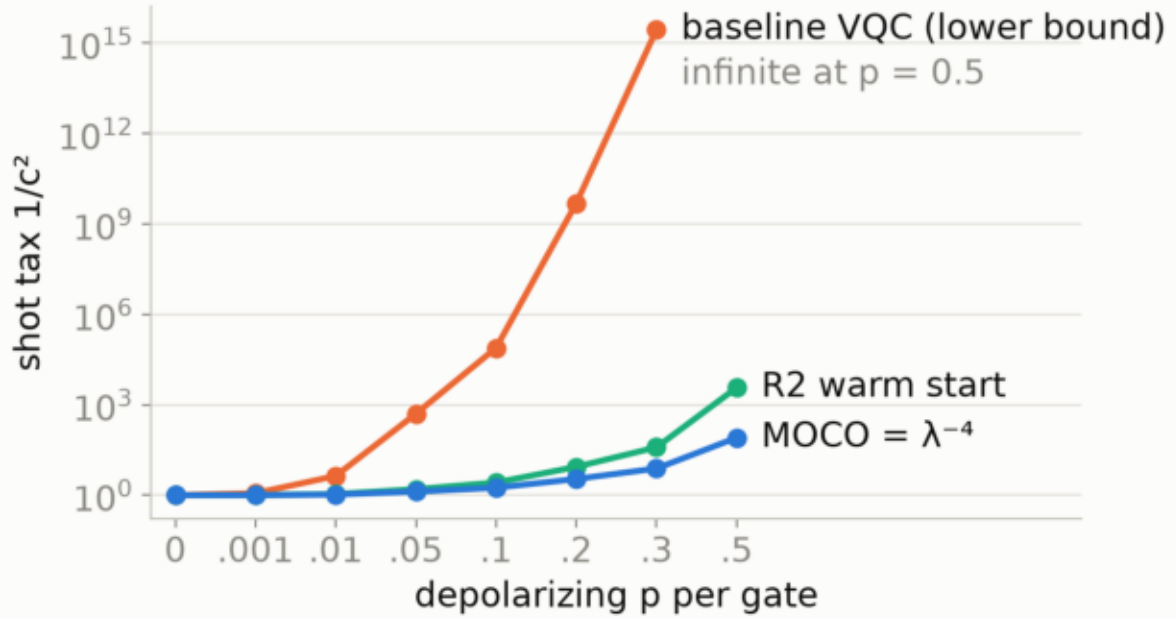
# With real shots noise is a tax: MOCO 1.32x, baseline at least 504x

**0.705**
MOCO, test AUPRC with 1024 shots at p = 0.05

**0.643**
R2 warm start, same setting

**0.007**
baseline: chance level



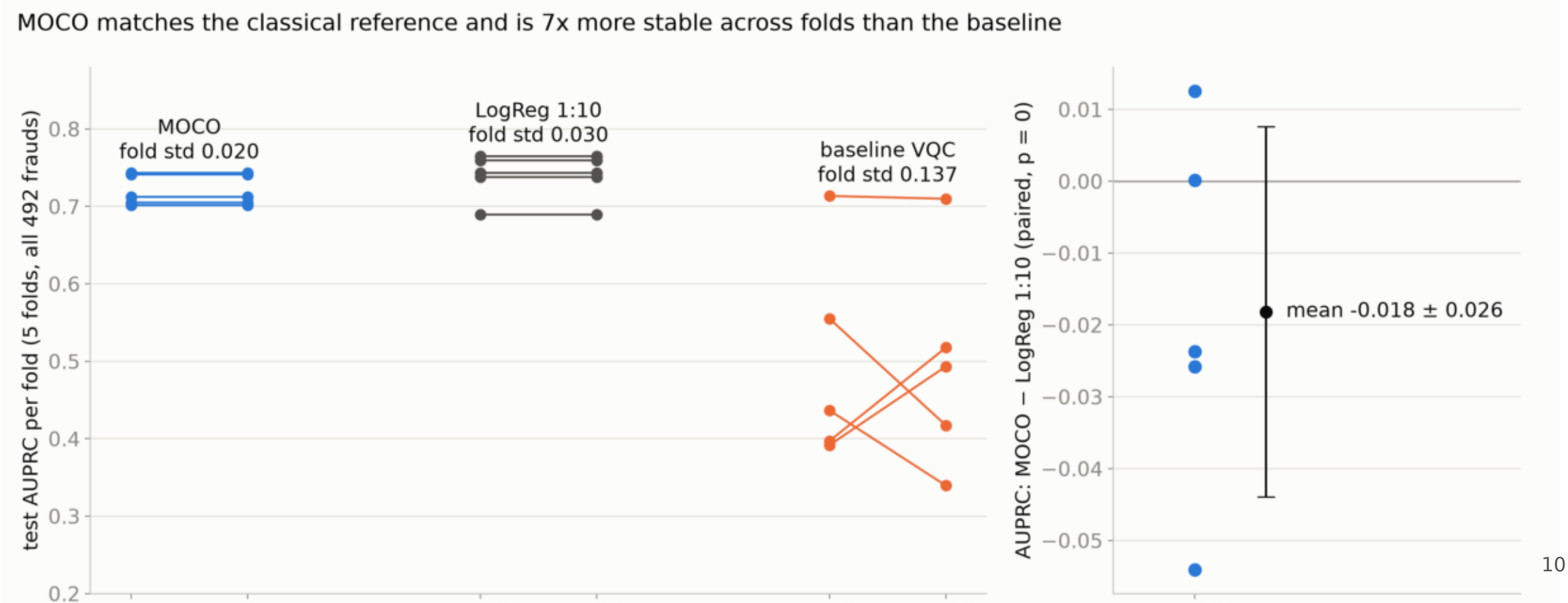

Shot tax $1/c^2$ at p = 0.05 (c = measured score compression): MOCO 1.32x (= $\lambda^{-4}$), R2 1.58x, baseline ≥ 504x (lower bound: its ranking is also lost).

# All 492 frauds: MOCO matches LogReg and is 7x more stable

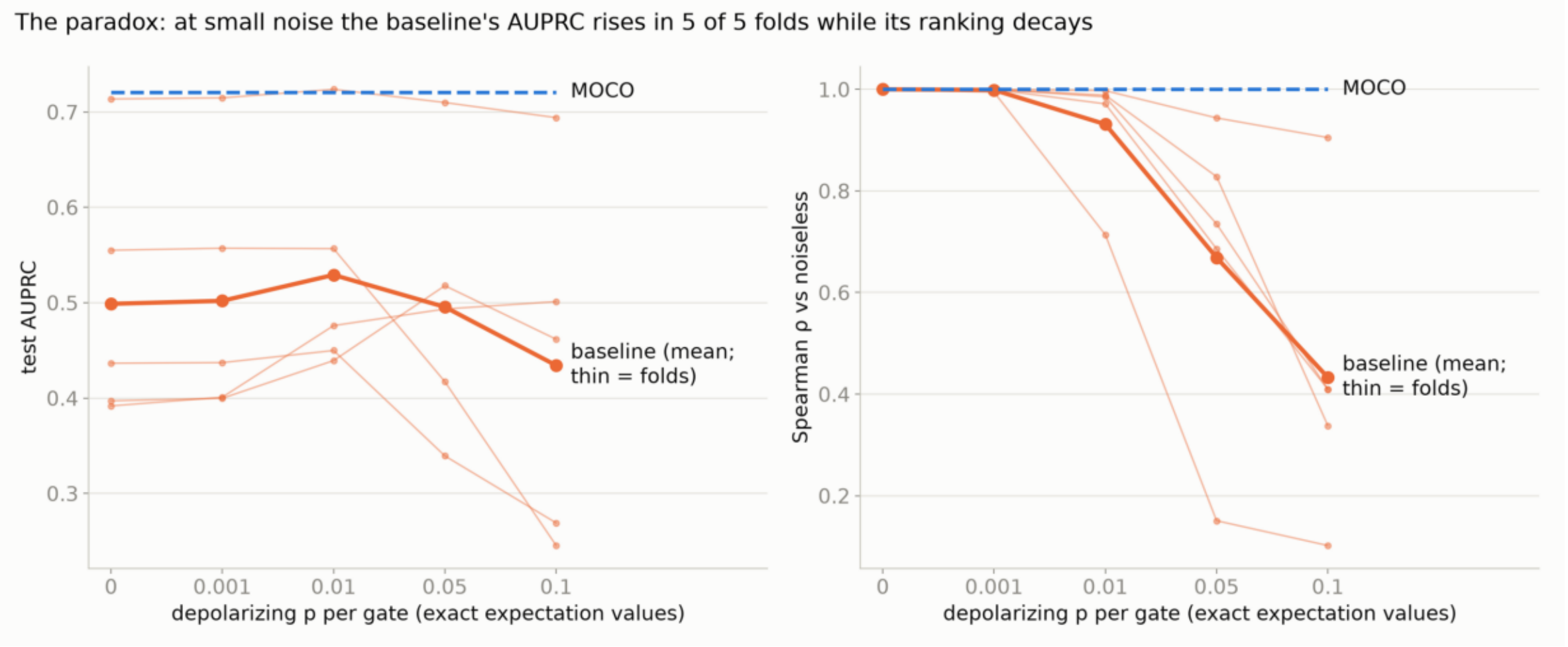
**0.721 ± 0.020**
MOCO, 5-fold CV, every p

**0.739 ± 0.030**
LogReg 1:10, same inputs

**0.499 ± 0.137**
baseline VQC, p = 0



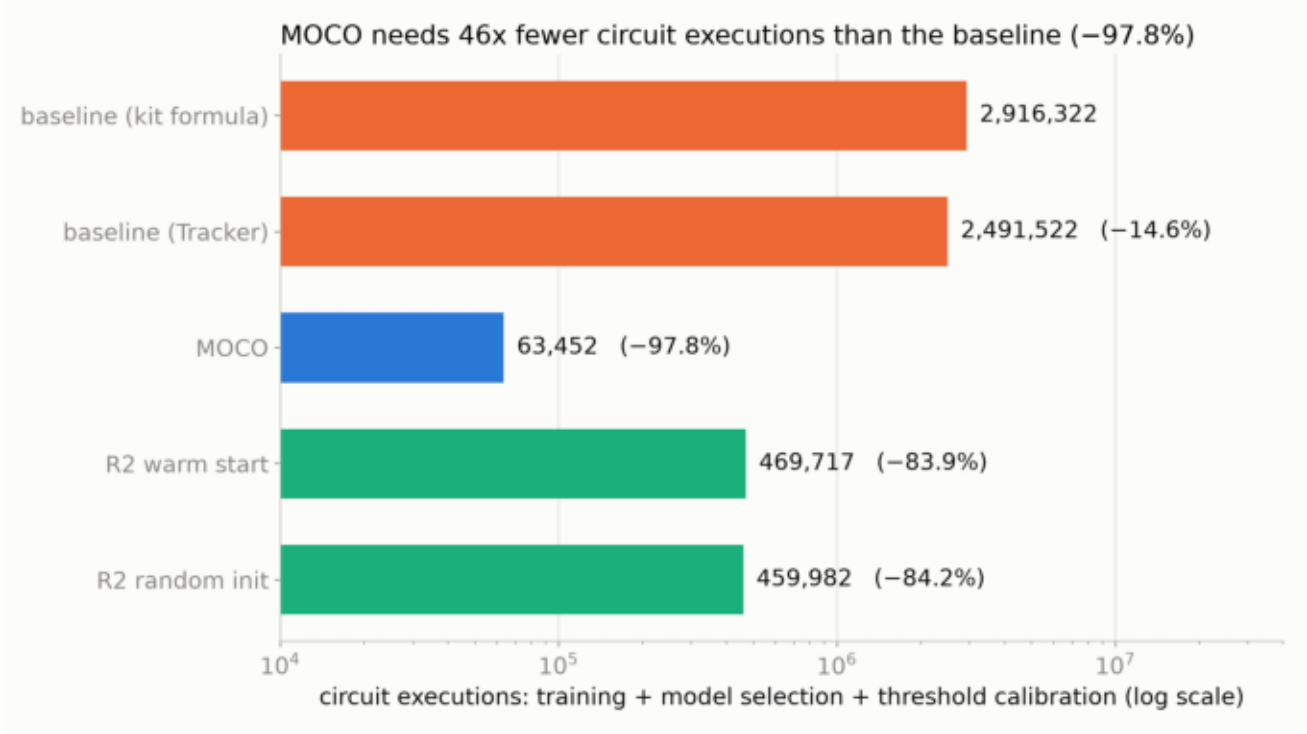
# Paradox: the baseline's AUPRC rises while its ranking decays




AUPRC rises at p = 0.01 in 5 of 5 folds while ρ falls (ρ = 0.67 at p = 0.05): one metric can report an improvement where the model gets worse, as in the starting article. MOCO cannot show this: its ranking is invariant.

# MOCO needs 46x fewer circuit executions

- Kit definition (training + model selection): baseline 2,859,360 vs MOCO 6,490 (-99.77%).
- With threshold calibration by the noise meter: baseline 2,916,322 vs MOCO 6,490 + 1,000 = 7,490 (-99.74%).
- Most conservative, threshold calibrated on the full noisy validation set: MOCO 63,452 (-97.8%).
- R2 warm start: 469,717 (-83.9%).



# Adaptive triage: 80% fewer shots at the same F1

## Plain intervals barely help

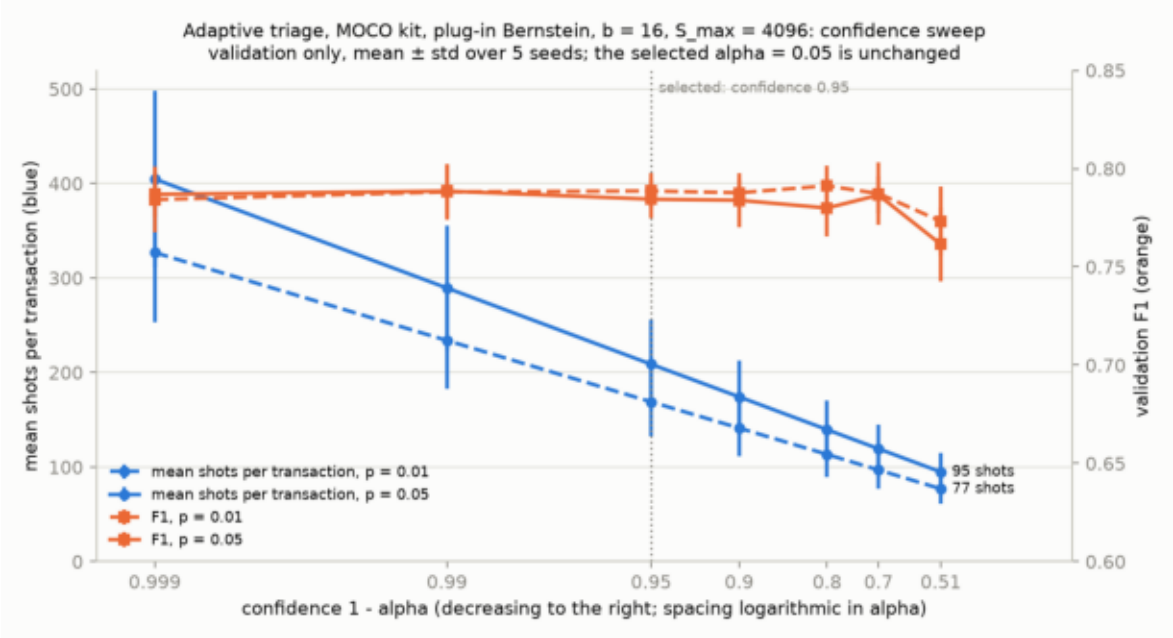
With Hoeffding-type (Maurer-Pontil) intervals, the kit encoding needs 992 shots per transaction on validation for the same F1: almost the fixed 1024.

## Selected on validation: plug-in Bernstein

b = 16, α = 0.05, S_max = 4096. Test, p = 0.05: 208 ± 47 shots per transaction, F1 0.801 ± 0.018 (fixed 1024: 0.797); 0 wrong early decisions.

## Noise meter

p estimated from 1,000 calibration executions instead of 56,962 (relative error 2.4% at p = 0.05); threshold moved analytically, F1 drop ≤ 0.0002.



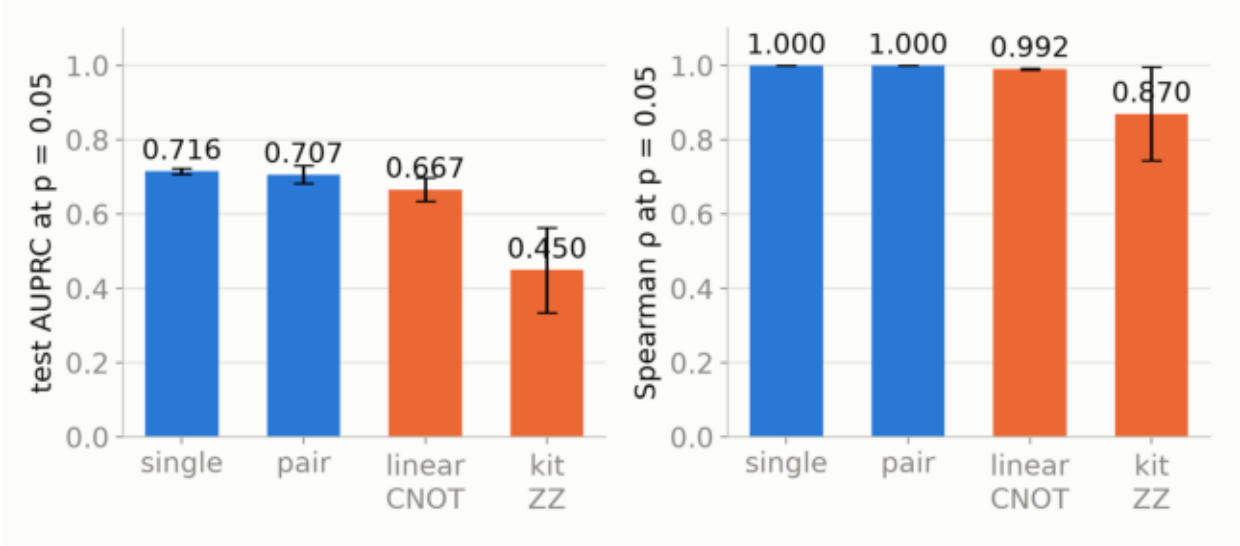

**Confidence 0.999 → 0.51: 4.3x fewer shots, F1 unchanged down to 0.7, wrong early decisions ≤ 0.0002; α = 0.05 fixed before the sweep**

Same fixed budget (209 shots for everyone): F1 0.702, so the adaptive rule spends shots where they matter. The baseline hits the 4096 cap for 99.7% of transactions (F1 0.09).

*Comparator, not selected: robust encoding q = 0.5% has protocol AUPRC 0.738 ± 0.006, LogReg on the same inputs 0.731 ± 0.004; validation preferred the kit encoding.*

# What did not work, measured honestly

- Light-cone pruning (R1): 15 gates removed, scores identical to 1e-10: zero gain.
- Entanglement: 3 CNOTs already lower AUPRC and break the invariance (figure).
- The kit's ZZ encoding caps at 0.44 even with the globally optimal readout.
- Noise-aware fine-tuning in R2: no gain over noiseless training.
- The shots-needed table was too noisy to use (validation and test agree in 24 of 40 cells).
- The baseline's drop under exact noise is fold-dependent (3 of 5 folds).



# What we claim, what we do not, what comes next

## We claim

- Robustness: $\rho = 1$ at every p, shot tax $\lambda^{-4}$.
- Stability: ±0.020 across folds (baseline ±0.137).
- Cost: 6,490 training executions (-99.77%).

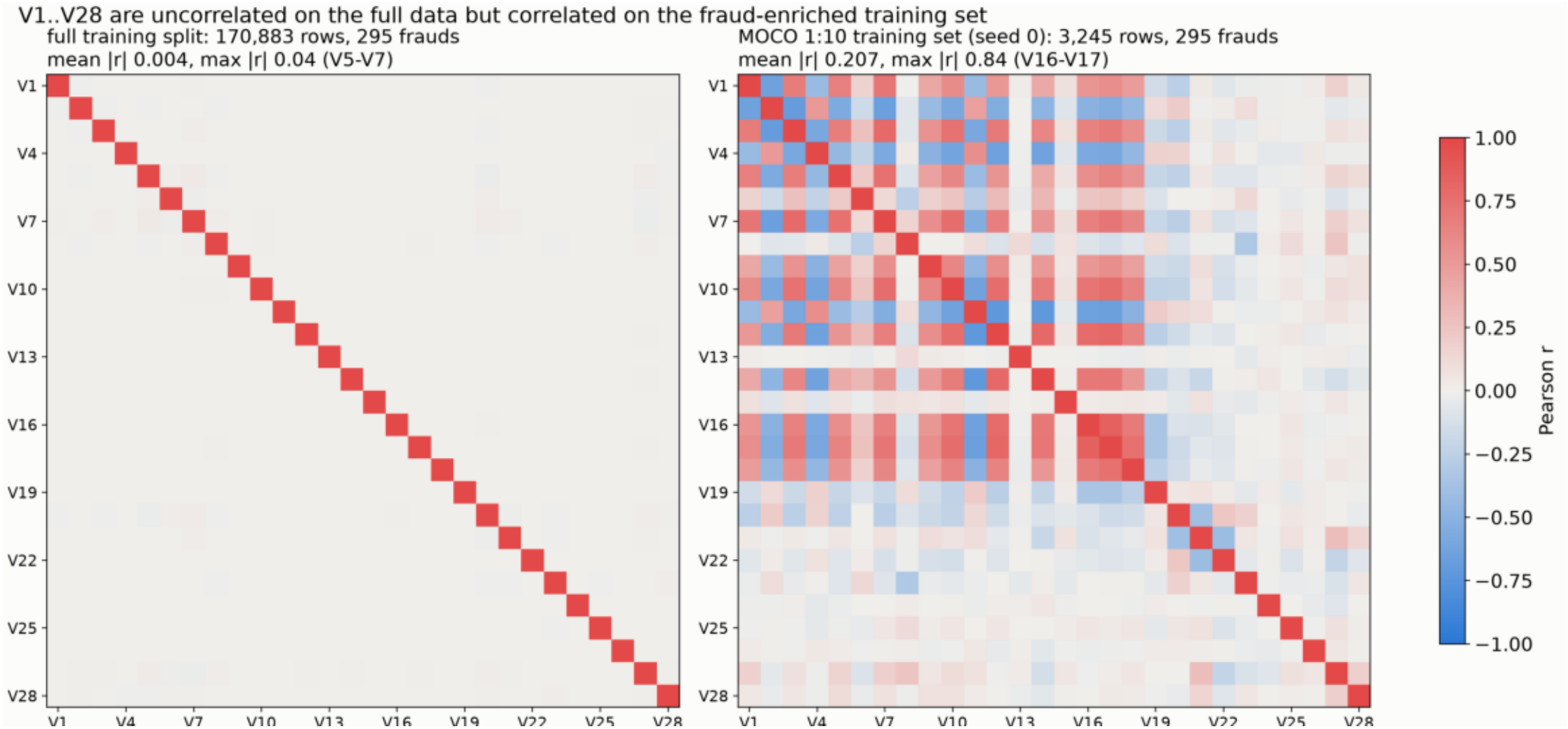
## We do not claim

- Quantum advantage: the circuit is a product state, classically simulable.
- Beating logistic regression: we match it (-0.018 ± 0.026 on all frauds).

**Next: real hardware noise (amplitude damping, readout error); more qubits at zero robustness cost.**

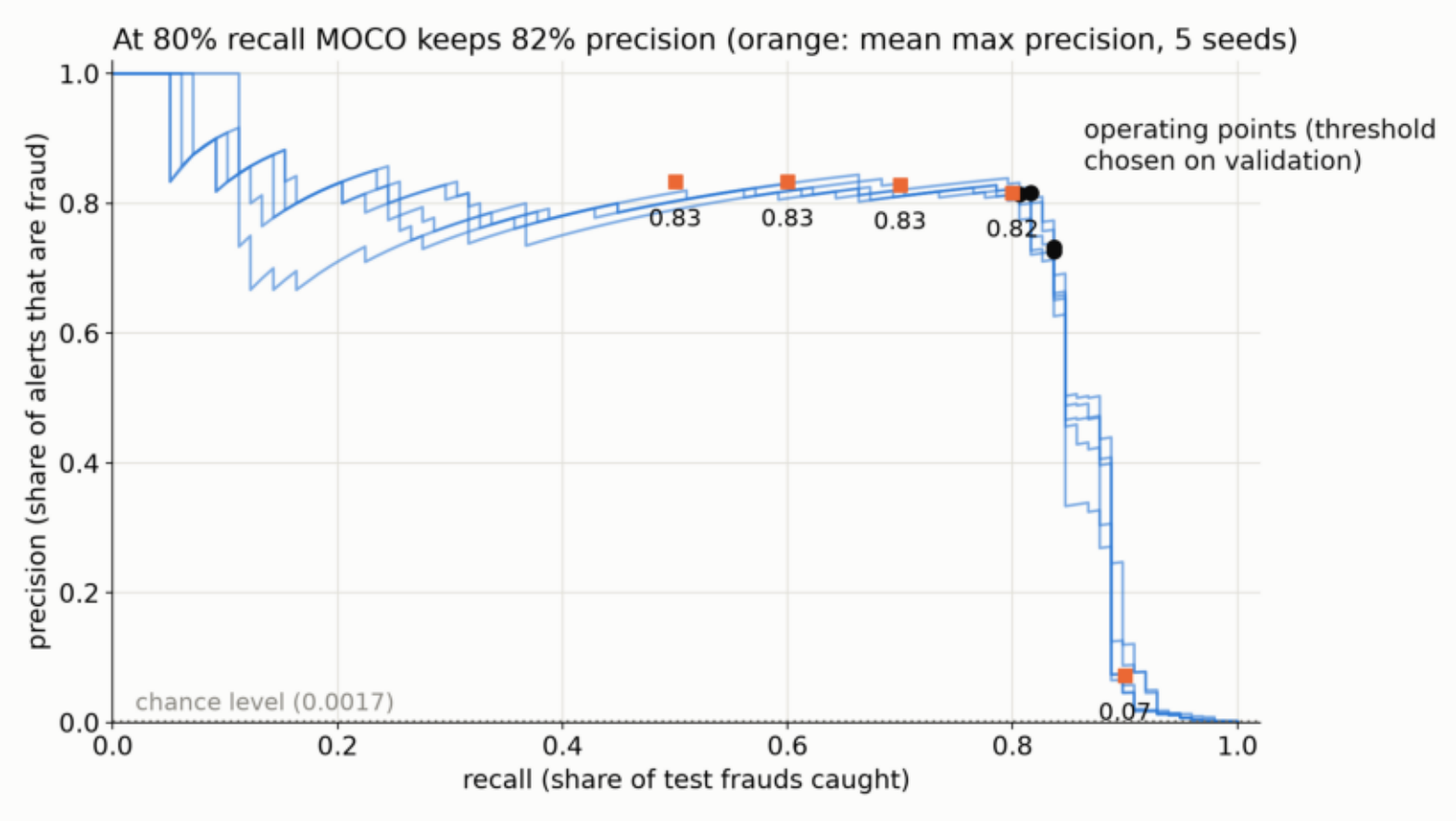
[1] Dinuț, Constantinescu, Alexandrescu, Electronics 15(11) 2489 (2026) · Schreiber, Eisert, Meyer, PRL 131, 100803 (2023) · Cerezo et al., Does provable absence of barren plateaus imply classical simulability?, arXiv:2312.09121 · Bowles, Ahmed, Schuld, arXiv:2403.07059 (2024) · Schuld et al., PRA 99, 032331 (2019) · Havlíček et al., Nature 567, 209 (2019) · Dal Pozzolo et al., IEEE SSCI (2015)

# Backup: features correlated on the 1:10 set, not on the full data




Full training split (170,883 rows): mean |r| 0.004, max 0.039. MOCO's 1:10 set (3,245 rows, 295 frauds): mean |r| 0.207, max 0.84 (V16-V17), 75 pairs above 0.5. The encoder's PCA is fitted on the 1:10 set.

# Backup: at 80% recall MOCO keeps 82% precision




| recall | max precision |
|---|---|
| 0.5 | 0.833 ± 0.008 |
| 0.6 | 0.833 ± 0.008 |
| 0.7 | 0.829 ± 0.007 |
| 0.8 | 0.817 ± 0.009 |
| 0.9 | 0.072 ± 0.032 |

Operating point (threshold chosen on validation): precision 0.781 ± 0.048, recall 0.822 ± 0.014; 81 frauds caught and 23 false alarms out of 98 test frauds (mean over 5 seeds).

*Test set, reporting only; no choice is made with it.*

# Backup: one fraud and one legit, end to end through MOCO

The validation transaction of each class with the median noiseless score (seed 0); the test set is not used.

## Validation fraud

| q | angle z | sin z | cos z | w_q | ⟨Z_q⟩ after Rot |
|---|---|---|---|---|---|
| 0 | 2.6319 | 0.4879 | -0.8729 | 0.4693 | 0.3727 |
| 1 | 2.6420 | 0.4791 | -0.8778 | 0.2160 | -0.9166 |
| 2 | 1.3252 | 0.9700 | 0.2431 | 0.2486 | 0.2859 |
| 3 | 0.7391 | 0.6736 | 0.7391 | 0.0661 | -0.9943 |

- ⟨H⟩ = Σ w_q ⟨Z_q⟩ = -0.0177, score (1 − ⟨H⟩)/2 = 0.5089
- p = 0: threshold t0 = 0.4510 → FRAUD
- p = 0.05: λ² = 0.8711, score 0.5077, transferred threshold 0.4573 → FRAUD

## Validation legit

| q | angle z | sin z | cos z | w_q | ⟨Z_q⟩ after Rot |
|---|---|---|---|---|---|
| 0 | 2.9434 | 0.1969 | -0.9804 | 0.4693 | 0.6392 |
| 1 | 2.2741 | 0.7627 | -0.6467 | 0.2160 | -0.7115 |
| 2 | 1.7583 | 0.9825 | -0.1864 | 0.2486 | 0.6617 |
| 3 | 1.2975 | 0.9629 | 0.2699 | 0.0661 | -0.7868 |

- ⟨H⟩ = Σ w_q ⟨Z_q⟩ = 0.2587, score (1 − ⟨H⟩)/2 = 0.3706
- p = 0: threshold t0 = 0.4510 → legit
- p = 0.05: λ² = 0.8711, score 0.3873, transferred threshold 0.4573 → legit

Checks: scores recomputed vs the cached protocol scores, all validation rows, max difference 2.8e-16; transferred threshold vs the one the protocol chose on noisy validation, difference 1.0e-14. Source: results/task2/e2e_example.md.